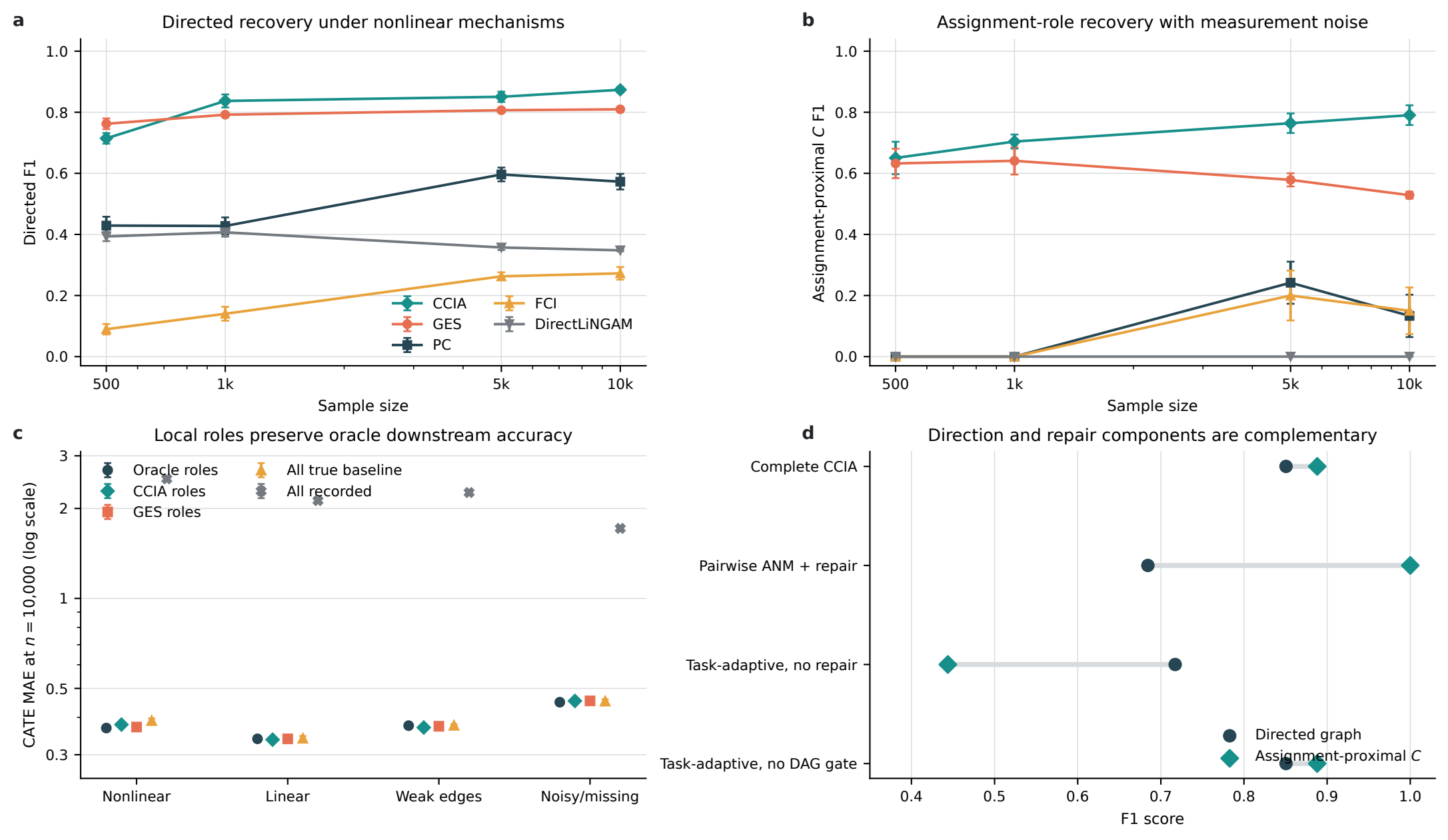
a
Directed recovery under nonlinear mechanisms
Directed F1
1.0
0.8
0.6
0.4
0.2
0.0
CCIA
GES
PC
FCI
DirectLiNGAM
500
1k
5k
10k
Sample size
b
Assignment-role recovery with measurement noise
Assignment-proximal C F1
1.0
0.8
0.6
0.4
0.2
0.0
500
1k
5k
10k
Sample size
c
Local roles preserve oracle downstream accuracy
CATE MAE at n = 10,000 (log scale)
3
2
1
0.5
0.3
Oracle roles
CCIA roles
GES roles
All true baseline
All recorded
Nonlinear
Linear
Weak edges
Noisy/missing
d
Direction and repair components are complementary
Complete CCIA
Pairwise ANM + repair
Task-adaptive, no repair
Task-adaptive, no DAG gate
Directed graph
Assignment-proximal C
0.4
0.5
0.6
0.7
0.8
0.9
1.0
F1 score